

Supplementary Material: An Obstruction Calculus for Viability under Incomplete Observation

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This supplement collects the material condensed from the main text: the complete proofs of the Section 3 certificates (S1), the expanded sufficiency review (S2), the bounded constructions and scope remarks (S3), and three additional figures (S4). S1 collects the complete proofs of the main text’s statements — including the selector principle, the sparse common-action witness, and the soundness of the oracle-recourse certificate (stated with its display in the support-function form) — together with the extended discussions and worked instances of the main text’s Sections 3, 4, and 10, as the closing blocks of this section. Notation and numbering follow the main text.

S1. Complete proofs

Theorem 1 (finite-horizon soundness and completeness). An initial belief B_0 admits an observation-based policy that keeps every compatible trajectory in $\mathcal{V}$ for N steps if and only if $B_0 \in \mathcal{W}_N$.

Proof. ($\Rightarrow$) Induction on N . The case $N = 0$ is $B_0 \subseteq \mathcal{V}$, i.e. $B_0 \in \mathcal{W}_0$. If a policy is safe for N steps, its first action a is admissible at every compatible state, so $a \in U^B(B_0)$; for every possible observation y , the successor belief $\text{Post}(B_0, a, y)$ is safe for $N - 1$ steps under the same policy, so by induction $\text{Post}(B_0, a, y) \in \mathcal{W}_{N-1}$; hence $B_0 \in \text{Pre}(\mathcal{W}_{N-1}) = \mathcal{W}_N$.

($\Leftarrow$) Backward induction. If $B_k \in \mathcal{W}_{N-k} = \text{Pre}(\mathcal{W}_{N-k-1})$, choose the witness action $a_k \in U^B(B_k)$; after the observation y_k the successor belief $B_{k+1} = \text{Post}(B_k, a_k, y_k) \in \mathcal{W}_{N-k-1}$, and the recursion continues. By construction every reachable state over N steps lies in $\mathcal{V}$. $\square$

Theorem 2 (common-action obstruction). Let $B \subseteq \mathcal{V}$ be an information set of the specified observation structure (the safe-control correspondence $\mathcal{R}_{\mathcal{V}}$ is defined on $\mathcal{V}$, so the belief is taken inside it). Suppose the following hypotheses hold.

(H2.1) The selected action is held until the next observation (sampled review: no informative observation arrives before the next action must be selected, and the policy holds the selected action over the review interval — the institutional reading; without a positive dwell time the instantaneous statement below must be replaced by the tube-safety statement below).

(H2.2) Either the **common admissible action set** is empty,

$$U^B(B) = \bigcap_{x \in B} U(x) = \emptyset, \quad (\text{admissibility obstruction})$$

or the **common safe-action set** is empty while common admissibility holds,

$$\mathcal{R}_{\mathcal{V}}^B(B) := \bigcap_{x \in B} \mathcal{R}_{\mathcal{V}}(x) = \emptyset \quad \text{with} \quad U^B(B) \neq \emptyset, \quad (\text{safety obstruction}) \quad (1)$$

(H2.3) **Closed-loop existence of the local adverse selection (safety case).** The disturbance correspondence is lower semicontinuous, or locally constant near the boundary

point at which the obstruction fires, so that the local adverse selection of the proof is realized along the compatible trajectory; alternatively, the certificate is read against the convexified (relaxed) inclusion, which preserves the signed drift (for C^1 q the directional derivative is affine in the velocity).

Then, under hypotheses (H2.1)–(H2.3), $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$: every compatible state may be individually robustly viable, and the belief is nevertheless nonviable, because the state-specific safe actions are incompatible.

Proof. Let π be any observation-based policy, and let $a = \pi(B)$ be the action it selects at the information set B .

Admissibility case. If $a \notin U^B(B)$, then for some compatible state $x \in B$ the action is inadmissible ($a \notin U(x)$): at that state the selected action is not an admissible control, so the policy fails outright on the compatible branch from x — an obstruction independent of the safe set.

Safety case. Otherwise $a \in U^B(B)$. Because $\bigcap_{x \in B} \mathcal{R}_{\mathcal{V}}(x) = \emptyset$ while $\mathcal{R}_{\mathcal{V}}(x) = U(x)$ at interior points of $\mathcal{V}$ (no constraint is active there) and $a \in U(x)$ for every $x \in B$, the emptiness of the intersection must be caused by the boundary: there is a state $\bar{x} \in B \cap \partial\mathcal{V}$ with $a \notin \mathcal{R}_{\mathcal{V}}(\bar{x})$. By the defining inequality of the safe-control correspondence, there are an active constraint j (with $q_j(\bar{x}) = 0$) and a disturbance $\bar{d} \in D(\bar{x})$ such that

$$\nabla q_j(\bar{x}) \cdot f(\bar{x}, a, \bar{d}) < 0.$$

With $\eta = -\nabla q_j(\bar{x}) \cdot f(\bar{x}, a, \bar{d}) > 0$, define the adverse-selection correspondence $D_{\eta}(x) = \{d \in D(x) : \nabla q_j(x) \cdot f(x, a, d) \leq -\eta/2\}$ on a neighbourhood of $\bar{x}$; by continuity of the right-hand side in (x, d) and the closed graph of D (Section 2.1), it has nonempty compact values near $\bar{x}$ and admits a measurable local selection $d(x)$ with $d(\bar{x}) = \bar{d}$. Let the disturbance realize this selection on a small initial interval (its closed-loop realization is hypothesis (H2.3)'s content — D lower semicontinuous or locally constant near $\bar{x}$, or the convexified reading); then along the compatible trajectory from $\bar{x}$ (compatible, since $\bar{x} \in B$), q_j strictly decreases from $q_j(\bar{x}) = 0$ at rate at least $\eta/2$, so the trajectory leaves $\mathcal{V}$ within the first review period — before any informative observation can arrive, whichever action the policy takes. Since π was arbitrary, no observation-based policy keeps every compatible trajectory in $\mathcal{V}$: $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$. $\square$

Proposition 1 (uniform margin gives a tube obstruction at every review length). Let $B \subseteq \mathcal{V}$ be a compact information set with $U^B(B) \neq \emptyset$, f and ∇q_j continuous, and suppose there is $\eta > 0$ such that for every $a \in U^B(B)$ there exist a boundary state $x_a \in B \cap \partial\mathcal{V}$, an active constraint j with $q_j(x_a) = 0$, and a disturbance $d_a \in D(x_a)$ satisfying

$$\nabla q_j(x_a) \cdot f(x_a, a, d_a) \leq -\eta.$$

Under the closed-loop existence convention of Theorem 2, the tube-safe action set is then empty at every review length,

$$\mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset \quad \forall \Delta > 0,$$

and hence $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ under sampled review of any length.

Proof. Fix $a \in U^B(B)$ and the triple (x_a, j, d_a) . By continuity of $(x, u, d) \mapsto \nabla q_j(x) \cdot f(x, u, d)$ and the closed graph of D , the drift inequality holds with margin $-\eta/2$ on a neighbourhood of (x_a, a, d_a) , and a measurable local selection $d(x)$ realizes it. Along the trajectory from x_a under the held action a and this adverse selection, $q_j(x(t)) \leq q_j(x_a) - (\eta/2)t = -(\eta/2)t < 0$ for every small $t > 0$, so the trajectory leaves $\mathcal{V}$ in arbitrarily small positive time; hence $\text{Reach}_{[0, \Delta]}(B, a) \not\subseteq \mathcal{V}$ for every $\Delta > 0$, i.e. $a \notin \mathcal{A}_{\text{tube}}(B, \Delta)$. Since $a \in U^B(B)$ was arbitrary, $\mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset$ for every $\Delta > 0$, and the tube-safety form certifies $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$. $\square$

Proposition 2 (obstruction ladder). Under the standing regularity assumptions and the closed-loop existence convention of Theorem 2, the common response sets of Section 2.4 are nested:

$$\mathcal{A}_{\text{tube}}(B, \Delta) \subseteq \mathcal{R}_{\mathcal{V}}^B(B) \subseteq U^B(B) \quad \forall \Delta > 0,$$

and emptiness descends the ladder:

$$U^B(B) = \emptyset \implies \mathcal{R}_{\mathcal{V}}^B(B) = \emptyset \implies \mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset \implies B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V}).$$

Proof. The second inclusion is by definition, since $\mathcal{R}_{\mathcal{V}}(x) \subseteq U(x)$ for every x . For the first, fix $a \in \mathcal{A}_{\text{tube}}(B, \Delta)$, $x \in B$ with an active constraint j ($q_j(x) = 0$), and $d \in D(x)$. If $\nabla q_j(x) \cdot f(x, a, d) < 0$, then along the trajectory from x under the held action a and the constant disturbance d , $q_j(x(t)) = \nabla q_j(x) \cdot f(x, a, d)t + o(t) < 0$ for all small $t > 0$, so the trajectory leaves $\mathcal{V}$ in arbitrarily small time, contradicting $a \in \mathcal{A}_{\text{tube}}(B, \Delta)$. Hence $\nabla q_j(x) \cdot f(x, a, d) \geq 0$ for every $d \in D(x)$ and every active j , i.e. $a \in \mathcal{R}_{\mathcal{V}}(x)$ for every $x \in B$, so $a \in \mathcal{R}_{\mathcal{V}}^B(B)$. $\square$

Theorem 3 (delayed-information obstruction). Let B_0 be an initial information set. Suppose the following hypotheses hold.

(H3.1) No informative observation arrives before time $T_{\text{obs}} > 0$ — formally, there are branches that are **observation-equivalent on** $[0, T_{\text{obs}})$: they produce identical observation records throughout that interval, so the policy cannot distinguish them before T_{obs} ; between observations the information set evolves by the set-membership semantics of Section 2.3.

(H3.2) There exist a constraint function q of class C^1 (the class of Theorem 4’s proof) with $\mathcal{V} \subseteq \{q \geq 0\}$ and a constant $\varepsilon > 0$ such that, **for every implementable blind-window control** — every measurable $u(\cdot)$ on $[0, T_{\text{obs}})$ that is admissible at every compatible state throughout the blind window, $u(t) \in U^B(B_t)$ with B_t the set-membership belief propagated under $u(\cdot)$ — which is exactly the class of controls that an observation-based policy can realize before the first informative observation, where its actions are a fixed function of time — there are a compatible initial state $x^* \in B_0$ attaining $q(x^*) = \inf_{x \in B_0} q(x)$ (with B_0 compact and q lower semicontinuous; otherwise replace the infimum by $\inf q + \delta$ for an arbitrary δ -minimizer, let $\delta \downarrow 0$, and read the timing bound with $\inf q + \delta$) and an admissible disturbance realization such that, along the realized trajectory from x^* under that open-loop control, the record remains compatible and

$$D^+q(x(t); f(x(t), u(t), d(t))) \leq -\varepsilon \quad \text{while } q(x(t)) > 0, \quad (2)$$

(H3.3) The timing bound holds:

$$T_{\text{obs}} > \frac{\inf_{x \in B_0} q(x)}{\varepsilon}. \quad (3)$$

Then, under hypotheses (H3.1)–(H3.3), $B_0 \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$: information may be accurate but arrive too late.

Proof. Fix any observation-based policy. By (H3.1) its actions on $[0, T_{\text{obs}})$ are a fixed open-loop function of time — the record carries no informative observation before T_{obs} , so every observation-equivalent branch induces the same actions; if this function is ever inadmissible at a compatible state the policy fails outright under the Section 2.4 convention, and otherwise (H3.2), applied to this implementable open-loop control, supplies a compatible initial state x^* with $q(x^*) = \inf_{x \in B_0} q(x)$ and an admissible disturbance realization satisfying (2) along the trajectory from x^* under those actions, observation-equivalent on $[0, T_{\text{obs}})$ to the other compatible branches. For q of class C^1 — the class assumed in (H3.2) — the fundamental theorem of calculus along the absolutely continuous trajectory integrates (2) to

$$q(x(t)) \leq q(x^*) - \varepsilon t = \inf_{x \in B_0} q(x) - \varepsilon t$$

while $q(x(t)) > 0$, so the trajectory attains the violation convention of Theorem 4 — the strict inequality $q(x(t)) < 0$ — by every time strictly larger than $t^* = \inf_{x \in B_0} q(x)/\varepsilon$ (in particular $\inf\{t : q(x(t)) < 0\} \leq t^*$; the same argument with the δ -minimizer bounds the violation time by $(\inf q + \delta)/\varepsilon$), and t^* by (3) is strictly less than T_{obs} — the violation is strict and precedes the first informative observation. Until t^* , the record contains no informative observation: the violating branch is observation-equivalent to every other compatible branch before T_{obs} , so the policy cannot have altered its action in response to the violation before it occurs. Since the policy was arbitrary and the violating realization is compatible with the observation record, no observation-based policy is robustly viable: $B_0 \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$. $\square$

Theorem 4 (finite-time exit certificate). Let $q : X \rightarrow \mathbb{R}$ be a constraint function of class C^1 — the class assumed throughout the exit theorems — with $\mathcal{V} \subseteq \{q \geq 0\}$, and suppose the following hypotheses hold.

(H5.1) There exist constants $a > 0$ and $\varepsilon > 0$ such that on the strip $\mathcal{S}_a = \{x : 0 \leq q(x) \leq a\}$,

$$\sup_{u \in U(x)} \inf_{d \in D(x)} D^+q(x; f(x, u, d)) \leq -\varepsilon \quad \forall x \in \mathcal{S}_a, \quad (4)$$

where $D^+q(x; v) = \limsup_{h \downarrow 0} [q(x + hv) - q(x)]/h$ is the upper right Dini derivative of q in direction v (for C^1 q , $D^+q(x; v) = \nabla q(x) \cdot v$ — the form the proof uses).

(H5.2) **Closed-loop existence of the adverse realization.** The disturbance correspondence is lower semicontinuous, or constant, on the strip, so that the state-dependent adverse selection constructed in the proof is realized by an admissible control–disturbance pair for which the Carathéodory existence theorem applies. (Alternatively, without (H5.2), the certificate is read against the convexified (relaxed) inclusion $\dot{x} \in \text{co } f(x, u(t), D_\varepsilon(x, u(t)))$, which admits solutions under Marchaud-type conditions and preserves the drift inequality because, for C^1 q , the directional derivative $\nabla q(x) \cdot v$ is affine in the velocity; closure and relaxation are the standard tools for this step.)

Then, under hypotheses (H5.1)–(H5.2), for every admissible control and every initial state $x_0 \in \mathcal{S}_a$ there exists an admissible disturbance realization under which the trajectory attains $q(x(t)) < 0$ — hence leaves $\{q \geq 0\} \supseteq \mathcal{V}$ — by every time strictly larger than $q(x_0)/\varepsilon$: $\inf\{t : q(x(t)) < 0\} \leq q(x_0)/\varepsilon$, in particular within time at most a/ε (since $q(x_0) \leq a$). (Convention, used in both exit theorems: *violation* is the strict inequality $q < 0$; at $q = 0$ the state may still lie on $\partial\mathcal{V} \subseteq \mathcal{V}$.)

Proof. Trajectories of $\dot{x} = f(x, u(t), d(t))$ exist in the Carathéodory sense under the standing hypotheses — D compact-valued, f measurable in t and continuous in (x, u, d) , linearly bounded — for every measurable control–disturbance pair. The adverse realization selected below is *state-dependent*, so its closed-loop realization is an additional requirement — the content of (H5.2) (D lower semicontinuous or constant, or the convexified reading of its parenthetical); convexity of D was not needed for the closure step — closure and relaxation are the standard tools for it. Fix an admissible control $u(\cdot)$. For each $x \in \mathcal{S}_a$, condition (4) gives $\inf_{d \in D(x)} D^+q(x; f(x, u(t), d)) \leq -\varepsilon$. Define the adverse-selection correspondence

$$D_\varepsilon(x, u) = \{d \in D(x) : D^+q(x; f(x, u, d)) \leq -\varepsilon\},$$

which has nonempty values on $\mathcal{S}_a$ by (4). For q of class C^1 the Dini derivative reduces to $\nabla q(x) \cdot f(x, u, d)$, continuous in (x, u, d) ; with the closed graph and compact values of D (Section 2.1), D_ε then has closed graph and compact values, and the measurable-selection theorem (Aubin and Frankowska, 1990, Thm. 8.1.3) supplies a measurable selection $d(\cdot)$ with $d(t) \in D_\varepsilon(x(t), u(t))$ along the trajectory. (The proof is carried out for q of class C^1 , as assumed; the locally Lipschitz case requires a separate comparison argument for almost-everywhere-differentiable arcs and is not treated.) Along the realization $(u(\cdot), d(\cdot))$ from x_0 , the fundamental theorem of calculus along the absolutely continuous trajectory integrates the inequality to

$$q(x(t)) \leq q(x_0) - \varepsilon t$$

as long as $x(t)$ remains in $\mathcal{S}_a$. Hence $q \leq 0$ at a time not exceeding $q(x_0)/\varepsilon \leq a/\varepsilon$. If the trajectory has not already left the strip by time $t^* + \delta$ for some $\delta > 0$, the comparison continues to apply and gives $q(x(t^* + \delta)) \leq q(x_0) - \varepsilon(t^* + \delta) \leq -\varepsilon\delta < 0$ with $t^* = q(x_0)/\varepsilon$; in either case the trajectory violates $\{q \geq 0\} \supseteq \mathcal{V}$ before every time strictly larger than $q(x_0)/\varepsilon$. The selection $d(t) \in D_\varepsilon(x(t), u(t))$ is itself nonanticipative feedback of the pair (x, u) — a disturbance strategy in the Isaacs order, responding to the current state and control without anticipating future controls — which is what the certificate requires; no Elliott–Kalton upgrade to a closed-loop strategy is required for the stated claim. $\square$

Proposition 3 (epistemic emptiness by admissibility — a minimal construction). There exists a system with $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}}) = \mathcal{V} \neq \emptyset$ such that, for a non-injective observation map O , every admissible initial information state — the observation fibres $B_0 = O^{-1}(O(S_0))$, $S_0 \in \mathcal{V}$ — lies outside $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$: the epistemic kernel over the fibre-induced initial beliefs is empty. The mechanism is *admissibility* rather than a hidden mode: the state-dependent control set of the construction leaves a single common action at the merged belief — one that is outward-drifting at the boundary state — so the emptiness is the common-action obstruction (Theorem 2’s safety case) at minimal size, produced by the control-set primitive rather than by information; the hidden-mode instance is Example 1.

Proof (by construction). Let $X = [1, 2]$, $\dot{S} = u - r(S)$ with $r : [1, 2] \rightarrow \mathbb{R}_{++}$ continuous and **injective on** $[1, 2]$ (e.g. $r(S) = S - 1 + c$, $c > 0$, strictly increasing), $U(S) = \{0, r(S)\}$, $\mathcal{V} = [1, 2]$, and the constant observation $O(S) \equiv 0$. Under perfect information, the state-feedback control $u = r(S)$ gives $\dot{S} = 0$, so every $S_0 \in \mathcal{V}$ is viable: $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}}) = \mathcal{V}$. Under the constant observation, the initial information state is the fibre $B_0 = [1, 2]$, and the set-membership semantics propagates it: at time t the belief is $B_t = \Phi_t(B_0) \cap \mathcal{V}$, the image of the initial set under the flow generated by the applied control — under a constant observation with no exit observation, this flow-image coincides with the record-compatibility definition of Section 2.3. The common control set at the initial belief is

$$U^B(B_0) = \bigcap_{S \in [1, 2]} \{0, r(S)\} = \{0\},$$

because $r(S) > 0$ for every S and, by injectivity on $[1, 2]$, no single control equals $r(S)$ for all S . An observation-based policy may select time-varying actions, but robust admissibility requires the selected action to lie in $U^B(B_t)$ at every t ; the propagated belief B_t is nonempty and, while the state remains in $[1, 2]$, $U^B(B_t) = \{0\}$ by the same computation (the flow of $\dot{S} = -r(S)$ with $r > 0$ moves B_t monotonically downward, preserving the injectivity of r on the surviving interval). Hence the policy must hold $u(t) = 0$ almost everywhere, giving $\dot{S} = -r(S) \leq -\min_{[1, 2]} r < 0$; every compatible trajectory exits $\mathcal{V}$ downward in finite time. The same argument applies verbatim to any initial fibre, since O is constant. Hence $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$ contains no fibre-induced initial belief, while beliefs that are singletons (exact prior knowledge) are not admissible initial states of this observation structure. The system has no disturbance and no estimation error: the failure is caused entirely by the non-injectivity of O . $\square$

Proposition 4 (observation-fibre criterion). An exact certifier based on O exists if and only if membership in K is constant on every observation fibre:

$$O(z_1) = O(z_2) \implies [z_1 \in K \iff z_2 \in K] \quad \forall z_1, z_2 \in Z, \quad (5)$$

equivalently, $K = O^{-1}(O(K))$ on the admissible domain.

Proof. ($\implies$) If C exists and $O(z_1) = O(z_2)$, then $\mathbf{1}_K(z_1) = C(O(z_1)) = C(O(z_2)) = \mathbf{1}_K(z_2)$, so membership is fibre-constant.

($\impliedby$) If (5) holds, define $C(y) = \mathbf{1}_K(z)$ for any $z \in Z$ with $O(z) = y$; (5) makes C well defined on $O(Z)$ and exact by construction; extend C arbitrarily on $Y \setminus O(Z)$. A *measurable* exact

certifier exists under the same condition on standard Borel spaces when $O(K)$ is measurable, taking $C = \mathbf{1}_{O(K)}$; measurability is the only additional requirement. $\square$

Corollary 1 (safety-crossing fibres and the certainly-safe set). If two admissible states share an observation and lie on opposite sides of the *same* component safety constraint — with the safe set of that certification problem read as the single floor $K = \{q_j \geq 0\}$; or, in the general form of the hypothesis, one of the two states lies in K and the other outside K — no exact observation-only certificate exists. (Crossing one floor of an intersection $K = \bigcap_j \{q_j \geq 0\}$ does not by itself put the pair on opposite sides of the intersection — the state satisfying floor j may violate floor i — hence the single-floor scoping.) The largest set of observations that can soundly be labeled safe without further information is the certainly-safe set

$$\mathcal{Y}_{\text{safe}} = \{y \in O(Z) : O^{-1}(y) \subseteq K\}.$$

Proof. In the general form (one state in K , one outside), the fibre violates (5) for that K , so Proposition 4 denies the certifier; in the single-floor form it violates (5) for the floor’s own certification problem ($K = \{q_j \geq 0\}$). On $\mathcal{Y}_{\text{safe}}$ every compatible state is safe, so the constant verdict “safe” is sound; outside $\mathcal{Y}_{\text{safe}}$ some compatible state is unsafe, so no sound “safe” verdict exists there. $\square$

Proposition 5 (monotonicity of the epistemic kernel). The robust epistemic kernel is monotone in its constituents: enlarging the action sets, reducing the disturbance sets, refining the information structure, or enlarging the policy class never shrinks the kernel. Writing $\text{ERViab}^{\cdot}$ for the kernel with the indicated constituent, $U_1 \subseteq U_2$, $D_1 \subseteq D_2$, $\Pi_1 \subseteq \Pi_2$, and “ $\mathcal{I}_1$ refines $\mathcal{I}_2$ ” imply respectively

$$\begin{aligned} \text{ERViab}^{U_1} &\subseteq \text{ERViab}^{U_2}, & \text{ERViab}^{D_2} &\subseteq \text{ERViab}^{D_1}, \\ \text{ERViab}^{\Pi_1} &\subseteq \text{ERViab}^{\Pi_2}, & \text{ERViab}^{\mathcal{I}_2} &\subseteq \text{ERViab}^{\mathcal{I}_1}. \end{aligned}$$

Proof. Each inclusion is by restriction: a policy that is viable under the smaller action sets, the larger disturbance sets, the smaller policy class, or the coarser information structure remains admissible and nonanticipative under the corresponding enlargement or refinement, while the adversary’s options are unchanged or reduced, so the same policy witnesses viability. $\square$

Complete proof of the selector principle (main text, Section 8). Let π witness $B \in \text{ERViab}_{\mathcal{I}}(\mathcal{V})$. (i) *Admissibility.* The policy π is admissible by the definition of epistemic viability, so its review selection $a = \pi(B)$ satisfies $a \in U(x)$ for every compatible $x \in B$ — an admissible policy never selects an action outside the state-dependent control set at a reachable state — whence $a \in U^B(B)$. (ii) *Safety.* Every trajectory compatible with B and governed by π over the review interval $[t_k, t_k + \Delta]$ lies in $\mathcal{V}$; under the sampled-hold semantics the union of these segments over compatible initial states is exactly $\text{Reach}_{[0, \Delta]}(B, a)$, giving the inclusion $\text{Reach}_{[0, \Delta]}(B, a) \subseteq \mathcal{V}$. (iii) *Recursive viability.* Let $B' = \tau(B, a, y)$ be any post-observation belief reached at the next review. Every $x \in B'$ lies on a trajectory compatible with the observation record through $t_k + \Delta$ that stayed in $\mathcal{V}$ so far and is governed beyond $t_k + \Delta$ by the continuation of π ; that continuation is admissible, information-adapted, and keeps every such trajectory in $\mathcal{V}$ — otherwise π would fail to keep every compatible trajectory of B in $\mathcal{V}$ on $[0, \infty)$. Hence $\pi|_{B'}$ witnesses $B' \in \text{ERViab}_{\mathcal{I}}(\mathcal{V})$, which is (iii). Induction on the review index now shows that all four response correspondences — U^B , $\mathcal{R}_{\mathcal{V}}^B$, $\mathcal{A}_{\text{tube}}$ at the review length, and the recursive predecessor — are nonempty at every reachable belief, each containing π ’s review selection; by contraposition, a certificate that any one of them is empty at a reachable belief certifies nonviability. The finite-system converse is the backward induction of Theorem 1 applied to the review-time belief recursion. The continuous-time converse is the backward-recursion completion recorded in Section 6.5 of the main text; it is cited there and not re-derived.

Complete proof of the sparse common-action witness (main text, Section 3; Proposition prop:helly). The statement is read with the constraint functions q_j convex in addition to C^1 — the standing hypothesis of the barrier literature this section joins; the convexity is load-bearing and is discussed in the remark after the proof. Fix x and write the safe-control set $\mathcal{R}_V(x) = \{u \in U(x) : q_j(F(x, u, d)) \leq 0 \ \forall j, \ \forall d \in D(x)\}$. *Closed.* Let $u_n \rightarrow u$ with $u_n \in \mathcal{R}_V(x)$ witnessed by disturbances $d_n \in D(x)$; compactness of $D(x)$ passes to a subsequence $d_{n_k} \rightarrow d \in D(x)$, continuity of F and of q_j gives $q_j(F(x, u, d)) = \lim_k q_j(F(x, u_{n_k}, d_{n_k})) \leq 0$, so $u \in \mathcal{R}_V(x)$. *Bounded.* As a subset of the compact U . *Convex.* For $u_1, u_2 \in \mathcal{R}_V(x)$, $\theta \in [0, 1]$, and any $d \in D(x)$: affinity of F in the control gives $F(x, \theta u_1 + (1 - \theta)u_2, d) = \theta F(x, u_1, d) + (1 - \theta)F(x, u_2, d)$, and convexity of q_j gives $q_j(\theta F(x, u_1, d) + (1 - \theta)F(x, u_2, d)) \leq \theta q_j(F(x, u_1, d)) + (1 - \theta)q_j(F(x, u_2, d)) \leq 0$; as d was arbitrary, the convex combination is safe. Now suppose $\bigcap_{x \in B} \mathcal{R}_V(x) = \emptyset$. Each member of the family $\{\mathcal{R}_V(x)\}_{x \in B}$ is compact and convex in the Euclidean space $U \subseteq \mathbb{R}^m$. If every finite subfamily had nonempty intersection, the finite-intersection property of the compact sets would give nonempty total intersection — contradiction; so some finite subfamily $\mathcal{R}_V(x_1), \dots, \mathcal{R}_V(x_N)$ has empty intersection. By Helly’s theorem in $\mathbb{R}^m$ (Helly, 1923), a finite family of convex sets in $\mathbb{R}^m$ with empty intersection contains a subfamily of at most $m + 1$ members with empty intersection; the corresponding $x_1, \dots, x_\ell$, $\ell \leq m + 1$, are the required witness. The converse direction is immediate — the witness sets are members of the family, so their empty intersection implies the full one.

Remark (on the convexity hypothesis). Without convexity of the q_j the safe-control set $\mathcal{R}_V(x)$ need not be convex — the nonconvex rows example of Section 3.4 of the main text exhibits exactly this failure — and the Helly route does not apply; the witness bound then holds for the convexified (relaxed) safe-control sets, a weaker statement consistent with the reading of the relaxation in Section 3.

Complete proof of the soundness of the oracle-recourse certificate (main text, Section 3.7), with the display corrected to the support-function form. Write the stored facets as functions g_j with the safe set $\{g_j \geq 0\}$, so $\beta_\ell^d = g_j(\bar{x}_\theta^d(t))$ is the zero-control facet value at the label and the realized facet value along branch θ under signals (w, u_θ) — w the common window signal on $[0, \tau)$, u_θ the branch’s post-reveal signal — is, by variation of constants in the control-affine dynamics (exact for linear control entry; to first order with the remainder charged against the stored margin in the general control-affine case),

$$g_j(x_\theta^{u,d}(t)) = \beta_\ell^d + \int_0^\tau (k_\ell^d)^\top w + \int_\tau^t (k_\ell^d)^\top u_\theta,$$

the kernel k_ℓ^d exactly as defined in the main text. *Corrected certificate.* Define, for weights $\lambda_\ell \geq 0$, $\sum_\ell \lambda_\ell = 1$,

$$\Gamma_h(\lambda) = \inf_{d(\cdot)} \left\{ \sum_\ell \lambda_\ell \beta_\ell^d + \int_0^\tau h_U \left(\sum_\ell \lambda_\ell k_\ell^d \right) d\sigma + \sum_\ell \int_\tau^T h_U(\lambda_\ell k_\ell^d) d\sigma \right\},$$

with $h_U(g) := \max_{v \in U} g^\top v$ the support function. If $\Gamma_h(\lambda) < 0$, then $B_0 \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$: some stored facet is violated along some branch, for every policy in the stated class and every branch selection. *Proof.* Let $d^*(\cdot)$ realize the infimum (or an infimizing sequence; the argument is pointwise in d). Consider any admissible policy: its window signal w is one fixed measurable function (the blind-window reduction of Theorem 3), and its post-reveal branch signals u_θ are arbitrary measurable functions. The realized weighted facet sum at d^* is

$$\sum_\ell \lambda_\ell g_{j_\ell}(x_{\theta_\ell}(t_\ell)) = \sum_\ell \lambda_\ell \beta_\ell^{d^*} + \int_0^\tau \left(\sum_\ell \lambda_\ell k_\ell^{d^*} \right)^\top w + \sum_\ell \int_\tau^{t_\ell} (\lambda_\ell k_\ell^{d^*})^\top u_{\theta_\ell},$$

and each pairing is bounded above by the support function, $g^\top v \leq h_U(g)$ for $v \in U$. Hence the realized weighted facet sum is at most $\sum_\ell \lambda_\ell \beta_\ell^{d^*} + \int_0^\tau h_U(\sum_\ell \lambda_\ell k_\ell^{d^*}) + \sum_\ell \int_\tau^T h_U(\lambda_\ell k_\ell^{d^*}) < 0$. A strictly negative weighted sum of the facet values forces some facet value to be strictly negative: the facet is violated along its branch. Since the window signal w was arbitrary, every policy in the class fails; branch selection is immaterial because the sum ranges over all stored labels. $\square$

Remark (an attainment-special variant). A variant certificate replaces h_U by the minimum $\phi_U(g) = \min_{v \in U} g^\top v$ and the firing inequality by the opposite sign. That form is sound exactly when the realized control pairings attain their pointwise minima, i.e. when the policy’s signals coincide with the ϕ_U -extremal selection; a chaining argument in the realized pairings requires this attainment and does not hold for an arbitrary admissible signal. The pooled construction of the three-branch instance satisfies the attainment hypothesis in the window term — the stored weights pool the pre-observation kernel to zero, $\sum_\ell \lambda_\ell k_\ell^d \equiv 0$ on $[0, \tau)$, for which $\phi_U(0) = h_U(0) = 0$ and every signal attains the bound — and on the instance the two forms share the exact critical delay $\tau^* = 7/50$ (the main text evaluates $\Gamma_h(\tau) = 7/50 - \tau$). In general the sound certificate is the support-function form Γ_h above, proved here; the ϕ_U -form is retained as the attainment-special case.

Complete discussion of the finite-horizon obstruction tree and the one-step instance (main text, Section 3). Theorem 1 restated with its complete machinery: If $B_0 \notin \mathcal{W}_N$, the failure is certified by a finite **obstruction tree**: at each node (B, k) with $k \geq 1$, for every $a \in U^B(B)$ the adversary can choose a possible observation y with $\text{Post}(B, a, y) \notin \mathcal{W}_{k-1}$; the branches are the compatible observation-and-disturbance choices that force violation within N steps. This is the exact finite-horizon counterpart of the sufficient certificates of the following subsections. The recursion reads B_0 as the belief after the initial observation has been assimilated, and the tree alternates regulator and nature turns — the regulator selects an action, the adversary a compatible observation — the standard alternating structure of partial-information games.

A one-step instance. The discrete one-step abstraction of Example 1 — transition $z^+ = z + \theta u$ on the grid $z \in \{-1, 0, 1\}$, preserving the common-action mechanism — is the smallest obstruction tree. With $z \in \{-1, 0, 1\}$, $\theta \in \{-1, +1\}$, $u \in \{-1, +1\}$, transition $z^+ = z + \theta u$, safe set $\mathcal{V} = \{z \geq 0\}$, and observation $O(z, \theta) = z$, the belief $B_0 = \{(0, -1), (0, +1)\}$ has $U^B(B_0) = \{-1, +1\}$. Under $u = +1$ the branch $\theta = -1$ reaches $z^+ = -1$ (observed -1), so $\text{Post}(B_0, +1, -1) = \{(-1, -1)\} \notin \mathcal{W}_0$; under $u = -1$ the branch $\theta = +1$ reaches $z^+ = -1$. Hence $B_0 \notin \mathcal{W}_1$, and the obstruction tree is the root $(B_0, 1)$ with two adversary edges, $\theta = -1$ and $\theta = +1$, each terminating in the violation $z < 0$. This is Theorem 2 (Section 3.2) in the finite-horizon language; the tree is drawn in Figure S2.

Complete discussion of the common-action certificate’s Farkas form (main text, Section 3). Theorem 2 in polyhedral data: Condition (1) is checkable: it requires computing the intersection of the safe-control sets over the information set, which for polyhedral constraint and control sets is a finite linear program. When the intersection is nonempty, the common action is a candidate for viability — Theorem 2 is a *necessary* condition with a constructive witness of failure, not a sufficient condition for success. For polyhedral data the emptiness certificate is a Farkas pair: with safe actions given by finitely many affine inequalities $A_i u \leq b_i$ across compatible states, stacked as $Au \leq b$, the intersection is empty if and only if there exists $\mu \geq 0$ with $\mu^\top A = 0$ and $\mu^\top b < 0$ (Farkas, 1902; Gale, 1960) — a finite, independently checkable certificate of epistemic infeasibility, the control-space analogue of the material-substitution separation certificates in the assessment-separation analysis of Abaee (2026). The certificate is finite when the safe-action constraints over the belief admit a finite representation (finitely many constraints across the compatible states); for infinite beliefs a finite-witness reduction is not automatic — the Helly-type witness of Proposition prop:helly of the main text (Section 3.4)

supplies one under the convexity hypotheses stated in Section 3.4. The worked instance of this certificate — the two-floor pair with multiplier $\lambda = (1/2, 1/2)$ and margin $1/10$ — is carried in the worked certificates below.

Complete discussion of the obstruction ladder (main text, Section 3). The rungs of Proposition 2, itemized: **The obstruction ladder.** Each certificate is a sufficient condition for emptiness at one rung:

- *Admissibility obstruction* — $U^B(B) = \emptyset$: no single action is admissible at every compatible state (the base case of Proposition 3).
- *Common safe-action obstruction* — $\mathcal{R}_V^B(B) = \emptyset$ with $U^B(B) \neq \emptyset$ (the safety case of Theorem 2).
- *Tube obstruction* — $\mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset$ (the tube-safety form).
- *Recursive obstruction* — in the finite-horizon setting the N -step common safe-action sets $\mathcal{A}_N(B)$ of Section 3.1 complete the ladder $\mathcal{A}_N(B) \subseteq \mathcal{A}_{\text{tube}}(B, \Delta) \subseteq \mathcal{R}_V^B(B) \subseteq U^B(B)$.

The rungs are strict in general: $\mathcal{R}_V^B(B) \neq \emptyset$ does not imply $\mathcal{A}_{\text{tube}}(B, \Delta) \neq \emptyset$ for $\Delta > 0$, since an instantaneously tangent action may still exit in finite time — the gap that Proposition 1 closes under a uniform margin. The dynamic exit certificate of Theorem 4 sits below the ladder: it empties the response set at some compatible state even under full information, which is the strongest form of obstruction and needs no observation-theoretic argument; the ladder, and the position of the exit certificate below it, are drawn in Figure S1.

Complete discussion of the delayed-information hypothesis and the timing bound (main text, Section 3). Theorem 3’s hypotheses, discussed: The hypothesis (2) is the set-membership form of the drift condition, quantified over open-loop controls on the blind window — a condition on the data of the problem, not on the policies: for every such control there is a compatible state and disturbance enforcing the drift; the adversary selects the true state and the disturbance, and the observations cannot expose the selection in time. It is this open-loop form that gives the timing bound (3) its force — (H3.1) converts the policy’s actions into one open-loop control, (H3.2) supplies the drift against it, and (3) then makes the obstruction epistemic rather than dynamic. The hypothesis is checkable in its special cases (constant observations; hold-until- T_{obs} policies) and is a template to be instantiated in the general case. Condition (3) is the **timing bound**: it states that the observation arrives *after* the enforced exit time — the cause of the obstruction; inverted, it reads as the design requirement that the first informative observation precede the enforced exit time $\inf q/\varepsilon$. Theorem 3 complements Theorem 2 quantitatively: the common-action obstruction is a **formal zero-margin counterpart** of the delayed tube-safety obstruction, and the two certificates remain distinct — the one a local action-feasibility condition, the other a finite-time survival-value condition — so the pair is not related as a $T_{\text{obs}} \rightarrow \infty$ limit. It is the obstruction that governs discrete-review governance: a review interval longer than the worst-case exit time is an information structure in which the exit is inevitable.

Complete discussion of the minimal admissibility construction (main text, Section 3). Proposition 3’s mechanism, isolated: The construction isolates an admissibility mechanism: the observation merges states whose *admissible* controls differ — the merged belief admits a common admissible control (the singleton $\{0\}$) but no common *safe* control, because that singleton is outward-drifting at the boundary state $S = 1$. The “inadmissible action = failure” convention is used explicitly: an action outside $U(x)$ at a compatible state x is itself

failure, and the state-dependent control set $U(S) = \{0, r(S)\}$ is a modelling primitive carrying that convention (Theorem 2’s admissibility case uses the same convention, stated there). The example also shows the empty-kernel phenomenon at its minimal size — two control values, one scalar state — as the minimal instance of Theorem 2 rather than a mechanism of equal rank; with a constant control set $[0, \bar{u}]$ in the same plant the fibre is epistemically viable (a constant control $u = r(m)$, $m \in (1, 2)$, drives every state of the fibre to the rest point $m \in (1, 2)$), so the construction is an admissibility mechanism rather than a hidden-mode mechanism. Theorem 2 states the general obstruction, of which this construction is the minimal instance.

Complete discussion of the fibre criterion’s governance reading (main text, Section 4). Corollary 1 in application: The fibre criterion has a direct governance reading. A composite sustainability index is an observation map from the state of a socio-ecological system to a scalar; a floor is a safe set. Corollary 1 then states: an index whose fibres cross a floor boundary — whose value cannot distinguish a state violating the floor from one satisfying it — cannot support an exact certification of that floor, however carefully the index is thresholded. The structural requirement is not that each floor be measured separately, but that the observation *separate every admissible pair lying on opposite sides of a specified floor*; direct per-floor measurement is one sufficient design, not the unique one, and any statistic refining the floor’s safe/unsafe partition certifies equally well. The certainly-safe set is the sound relaxation: it is the set of observation values where the index may certify safety; outside it, the index must fall silent.

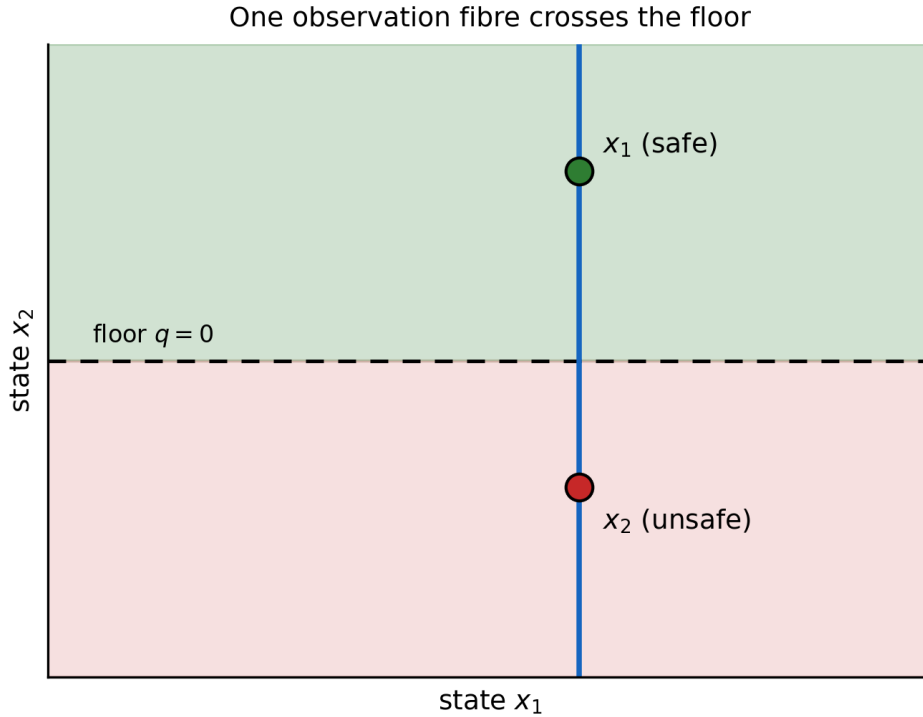


Figure 1: An observation fibre crossing a floor boundary. The observation $O(x) = x_1$ cannot separate the safe state $(0.6, 0.8)$ from the unsafe state $(0.6, 0.3)$ on the same fibre, so no exact observation-only certifier exists (Proposition 4).

Complete discussion of the degenerate (deterministic) limit: the worked instances and the audit-slice table (main text, Section 10). The deficit bound of the degenerate

z_0	T_{obs}	V_1	V_2	V_3	audit verdict	covering certificate
1.5	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	•	common-action
1.5	2	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	•	common-action
1.5	3	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	•	common-action
2.0	1	1	1	1	◦ viable	—
2.0	2	1	$\frac{1}{2}$	$\frac{1}{2}$	*	timing bound
2.0	3	1	$\frac{1}{2}$	$\frac{1}{2}$	*	timing bound
2.5	1	1	1	1	◦ viable	—
2.5	2	1	$\frac{1}{2}$	$\frac{1}{2}$	*	timing bound
2.5	3	1	$\frac{1}{2}$	$\frac{1}{2}$	*	timing bound

Table 1: Belief-state safety values $V_k(b_0)$ on the audit slice, computed over the declared hold class (exact rational values; the script regenerates all 48 cells). The deficit is $\frac{1}{2}$ on every nonviable cell — the attained bound of the proposition (main text, Section 10) — and $V_k \equiv 1$ exactly on the viable region $z_0 \geq 1 + T_{\text{obs}}$.

limit, attained: The proposition’s deficit bound is attained on both worked instances: It is a finite-model statement; a continuous-state limit requires uniform concentration of the kernels, which is not developed here. The deficit from 1 is at least the smallest compatible mass $\min_{x \in B} b(x)$.

A worked instance (the two-floor conflict). Read the two-floor system of Section 3.4 as a finite POMDP: states x_1 and x_2 share one observation; the action $u \in [0, 1]$ is admissible; the one-step violation probability is $p(x_1, u) = 1$ for $u > 0.4$ and 0 otherwise, and $p(x_2, u) = 1$ for $u < 0.6$ and 0 otherwise; the prior is uniform, $b_0 = (1/2, 1/2)$. Then $\sum_x b_0(x) p(x, u) \geq 1/2$ for every u , so the chance-constrained proposition of the main text (Section 10) gives $V_k(b_0) \leq 1/2$ for all $k \geq 1$, and the chance constraint $\mathbb{P}_{b, \pi} \geq 1 - \varepsilon$ is infeasible for every $\varepsilon < 1/2$. The bound is attained: $V_1(b_0) = 1/2$ (each action sacrifices exactly one floor), and the surviving floor is thereafter kept safe by its own action, so $V_k(b_0) = 1/2$ for all k . At the degenerate limit this reads $B_0 = \{x_1, x_2\} \notin \mathcal{W}_1$ with $V_k(b_0) = 1 - \min_{x \in B_0} b_0(x) = 1/2$, the bound of the proposition (main text, Section 10) — the chance-constrained obstruction evaluated at its exact value.

A worked instance (the delayed hidden regime on the audit grid). Read the audit system of Section 9 of the main text as a finite POMDP over the declared hold class: states (z, θ) with $\theta \in \{-1, +1\}$, prior $b_0 = (\frac{1}{2}, \frac{1}{2})$ on the two regimes from z_0 , no observation before T_{obs} , exact regime revelation at T_{obs} , and blind-window actions restricted to the declared holds $u \equiv \pm 1$. The safety values are exact: for $k \leq T_{\text{obs}}$,

$$V_k(b_0) = \frac{1}{2}(\mathbf{1}[z_0 - k \geq 1] + \mathbf{1}[z_0 + k \geq 1]),$$

since a hold drives one branch down at unit rate; and for $k > T_{\text{obs}}$, $V_k(b_0) = \frac{1}{2}(1 + \mathbf{1}[z_0 \geq 1 + T_{\text{obs}}])$, the revealed regime then held to growth. Three alignments follow. First, $V_1(b_0) = 1$ exactly when $z_0 \geq 2$: the one-step certificate’s silent region is exactly the region where the first blind step is safe. Second, $V_k(b_0) = 1$ for every k exactly when $z_0 \geq 1 + T_{\text{obs}}$: the belief-state value reproduces the audit’s viability boundary, so on this grid the certificate partition and the value recursion agree cell by cell. Third, every deficit equals $\frac{1}{2} = \min_x b_0(x)$: on each nonviable cell the bound of the proposition (main text, Section 10) is attained — the declining branch is the lost mass, and the chance-constrained obstruction is again evaluated at its exact value. Table 1 tabulates the slice $z_0 \in \{1.5, 2, 2.5\}$; the verification script evaluates all 48 cells in exact rational arithmetic and regenerates the table verbatim. The class declaration is load-bearing here, as in clause (iv) of the class-declaration theorem of the main text (Section 10): over unrestricted time-varying blind policies the alternating sequence keeps both branches within $z_0 \pm 1$, so the unrestricted value equals 1 exactly on $\{z_0 \geq 2\}$ — the cells the timing bound covers are exactly the cells where in-window adaptivity rescues the declared hold class.

S2. The sufficiency landscape (full)

5. The Sufficiency Landscape

The sufficiency results against which the obstruction calculus is defined are standard; they are stated here for contrast, and proofs appear in the cited sources.

(a) Veliov’s output-feedback condition. Veliov (1993) considers the same problem as Section 3 (a tube in the state space under incomplete and inexact measurement) and gives a sufficient condition for the existence of an *output-feedback regulation map*: a set-valued feedback of the measurement under which all trajectories starting from the graph of the tube remain in it. Under perfect measurement his condition reduces to the classical viability condition (Haddad, 1981). The complementarity with this paper is exact: Veliov’s theorem certifies existence; the obstruction calculus certifies nonexistence; a problem that satisfies neither requires a stronger observer or a finer observation structure.

(b) The estimation-tube programme. Cardaliaguet, Quincampoix, and Saint-Pierre (2007) pass from imperfect information in the measurement space to perfect information in an estimation space of *sets* of compatible states: the value functions of the two problems coincide, and the estimation-space problem admits a Dini-derivative characterization. The reduction is exact for value functions; what it does not supply is a *common prescription*. The exact relationship is stated in Section 6.3: carried out under the set-membership semantics, the estimation-space solution propagates the belief under a single control signal, so its controls *are* common controls and joint admissibility is built into the estimation-space problem’s own admissibility. Pointwise selections taken without that joint-admissibility check need not be jointly admissible; the common-action obstruction (Theorem 2) is the certificate that no jointly admissible selection exists at the information state in question.

(c) Observer-and-buffer transfer. When a full-information feedback exists with a strict inward margin, and an observer supplies exponentially convergent estimates, output feedback of the estimate preserves safety on a buffered subset: if K_ζ is a compact controlled-invariant subset of the interior of the kernel, the state feedback $u = k(x)$ is Lipschitz with constant L_k , the observer satisfies $\|\hat{x}(t) - x(t)\| \leq Me^{-\lambda t}\|\hat{x}(0) - x(0)\|$, and the initial estimation error is small enough that $L_k M\|\hat{x}(0) - x(0)\|$ is absorbed by the margin, then K_ζ is viable under output feedback (observer-to-viability transfer with safety buffer; the standard separation argument of observer-based control (Sontag, 1998)). Similarly, eroded kernels: if K is robustly invariant under full-state feedback with a strict inward margin on ∂K , and estimation and implementation errors are bounded by ζ , then an eroded set $K^{-c\zeta}$ is invariant under output feedback for a sensitivity constant $c > 0$ (erosion absorbs the error; see the buffer constructions in the literature cited in Section 1.3). These results bound the sufficiency side: under margins and convergence, output feedback *can* work; the obstruction calculus of Sections 3–4 states the conditions under which it *cannot*.

(d) The linear substitution alternative. For the finite linear model in which a resource-typed system must meet a demand vector through specified substitution pathways, exactly one of the following holds (Farkas, 1902; Gale, 1960): (i) there exists a non-negative pathway vector $a \geq 0$ satisfying the linear substitution constraints; or (ii) there exist multipliers $\alpha, \beta, \gamma \geq 0$ such that

$$\alpha^\top R + \beta^\top E - \gamma^\top Q \geq 0 \quad \text{componentwise,} \quad \gamma^\top s^{\text{req}} > \alpha^\top x + \beta^\top e.$$

The second statement is a certificate that the specified substitution pathways cannot meet demand within the typed resource and capacity bounds; writing all constraints as $Aa \leq \rho$, the pair is the Farkas lemma alternative. The multipliers are a separation certificate, not universal exchange rates: nonlinear, nonconvex, path-dependent, spatial, or irreversible technologies require their own feasibility analysis, and an elasticity fitted near one operating point cannot establish global substitutability. This is the feasibility-side complement of the observation-side

obstructions of Sections 3–4: where the present paper’s certificates bound what *information* can achieve, the substitution alternative bounds what *material pathways* can achieve, and the same infeasibility logic — exhibit the separating multiplier rather than assert impossibility — applies to both.

A worked comparison: barrier, estimation-tube, and obstruction certificates on one system. The two-floor system of Section 3.4 makes the division of labour concrete. Two states x_1 and x_2 share one observation; the single control $u \in [0, 1]$ must satisfy $u \leq 0.4$ at x_1 (floor 1 active) and $u \geq 0.6$ at x_2 (floor 2 active).

Barrier certificates. A barrier certificate certifies safety from a scalar function whose zero level set separates the unsafe region from all trajectories under some state feedback. Here there is no state to feed back — the observation merges x_1 and x_2 — so an admissible barrier would have to be constant on the fibre $\{x_1, x_2\}$, and no such function can separate the two floors’ incompatible requirements. The programme therefore reports “no barrier found,” a verdict that does not distinguish “unsafe” from “unsafe for the class of barrier functions tried.”

Estimation-tube reduction. Lifting to the estimation space replaces the merged observation by perfect information about the belief $\{x_1, x_2\}$. The belief-state viability problem then has empty value — no common control keeps both floors safe — so the estimation-space kernel is empty. The reduction returns the verdict exactly, but as a black box: it does not isolate which feature of the observation is responsible.

Obstruction certificate. The common-action obstruction (Theorem 2) fires with the explicit Farkas certificate $\lambda = (1/2, 1/2)$ for the stacked system $u \leq 0.4$, $-u \leq -0.6$, with $\lambda^\top A = 0$ and $\lambda^\top b = -0.1 < 0$ — the infeasibility margin 0.1 of Section 3.4. The sparse witness of Section 3.4 records that $m + 1 = 2$ states already suffice, and the certificate licenses a concrete remedy: any observation separating x_1 from x_2 restores one-step viability. (The general polyhedral statement and its checkability discussion are the complete-discussion block of S1.) This is the division of labour the paper relies on — the barrier programme and the estimation-tube reduction certify *that* the kernel is empty; the obstruction calculus certifies *why*, and names the observation feature to change.

Complete proof of the linear-programming instantiation theorem (main text, Section 3.3). Unroll the affine dynamics: for step k ,

$$x_{j,k} = A_j^k x_{j,0} + \sum_{i=0}^{k-1} A_j^{k-1-i} (B_j u_i + E_j w_{j,i}), \quad w_{j,i} \in W_j.$$

The robust floor row $F_i x_{j,k} \leq g_i$ holds for every disturbance sequence if and only if

$$\sum_{i=0}^{k-1} (F_i A_j^{k-1-i} B_j) u_i \leq g_i - F_i A_j^k x_{j,0} - \sum_{\ell=0}^{k-1} h_{W_j} (E_j^\top (A_j^{k-1-\ell})^\top F_i^\top),$$

where h_{W_j} is the support function of W_j ; each support value is itself a linear program over the polytope W_j , precomputed once, exactly in rational arithmetic. The horizon- K' feasibility program is therefore $\{Gu \leq b\}$ in the variables $u_0, \dots, u_{K'-1}$, with one row per (branch, step, floor constraint) and the box rows of U ; its size is polynomial in K' , $|J|$, and the description lengths. Feasibility is monotone in K' because the rows of a shorter horizon are a subset of those of a longer one, so $\sigma^*(B_0; \Pi_B) = \max\{K' \leq K : \text{feasible}\}$ is computable by bisection with $\lceil \log_2 K \rceil + 1$ programs: claim (i). Claim (ii) is Farkas’ lemma: infeasibility of $\{Gu \leq b\}$ yields $\lambda \geq 0$ with $\lambda^\top G = 0$ and $\lambda^\top b < 0$; indexing λ by (branch, step, constraint) gives nonnegative adjoint rows whose aggregate control projection vanishes and whose aggregate

constant is negative. Claim (iii) is definitional: the program's feasible set is, by construction, $\{(u_0, \dots, u_{K-1}) \in \Pi_B : \text{every branch in } \mathcal{V} \text{ through step } K\}$, so no relaxation is taken and the computation is sound and complete over the declared class.

For claim (iv), the two computations. *Hidden-regime gap*: with branches $z^+ = z + u$, $z^- = z - u$, floor $z \geq 1$, and declared finite class $\{\pm 1\}$, any hold control drives one branch down at unit rate, which reaches the floor at time $z_0 - 1$: $\sigma^* = z_0 - 1$. Enlarging the declaration to the polytope $[-1, 1]$ admits $u \equiv 0$, under which both branches remain at $z_0 \geq 1$ forever: $\sigma_{\text{rel}}^* = \infty$, and the relaxed timing certificate never fires. The gap between the relaxed and implementable certificates is therefore unbounded, and soundness of the relaxed verdict does not restore completeness over the implementable class. *Duality computation*: with branches $z^+ = \frac{9}{10}z \pm u$, the level- k rows are $z_k^+ \geq 1$ and $z_k^- \geq 1$ with $z_k^\pm = (\frac{9}{10})^k z_0 \pm \sum_{i < k} (\frac{9}{10})^{k-1-i} u_i$. Summing the pair cancels the control terms pairwise (equal magnitudes, opposite signs), leaving $2(\frac{9}{10})^k z_0 \geq 2$: a Farkas combination proving $z_0 \geq (\frac{10}{9})^k$, with the vanishing aggregate control projection visible as the pairwise cancellation. Necessity also follows control-independently from the branch sum $z_k^+ + z_k^- = 2(\frac{9}{10})^k z_0$; sufficiency at the boundary $z_0 = (\frac{10}{9})^K$ is witnessed by $u \equiv 0$, which keeps the branches balanced at $(\frac{10}{9})^{K-k} \geq 1$. The threshold is therefore exact, and the program is a complete characterization of the declared class. All twelve computations in this proof and the next are machine-verified in exact rational arithmetic (fractions, no floating point): the survival-time identity and the $42 = 30 + 12$ audit partition of the hidden-regime grid; the extinguishing of the certificate under the polytope relaxation; the level- k pairwise cancellations and deepest bound for $K \leq 3$; the boundary control; and the three-state computations below.

Complete proof of the exact two-phase decomposition (main text, Section 7). Write a policy for the two-phase model as a blind part $(u_0, \dots, u_{K-1}) \in \Pi_B$ followed by an adapted part, a map from the revealed state to a continuation. (i) $\Rightarrow$ *branchwise necessity*: branches evolve independently under the blind control, and viability requires $\mathcal{V}$ -membership at every time in every realization, so a blind control under which some branch exits $\mathcal{V}$ before the reveal is not window-surviving. (ii): if some branch lands outside $\text{RViab}(\mathcal{V})$ at the reveal under every window-surviving blind control, then upon that branch's realization no adapted continuation is viable, and the policy fails; conversely, if some window-surviving blind control lands every branch in $\text{RViab}(\mathcal{V})$, the policy consisting of that blind part and, after the reveal, the kernel-witnessed continuations from the revealed state is viable: after the reveal the information is complete, and membership in the full-information kernel is exactly the existence of a viable continuation. This proves the equivalence. For the consequence, over a class on which the window certificates are complete, silence of the pair is equivalent to clause (i): each certificate fires only on witnessed window failure (common-action at one step; timing by exit before the horizon), so if (i) holds neither fires, and if (i) fails the complete window certificate fires. Given silence, viability holds exactly when clause (ii) holds for some window-surviving blind control, and nonviability, when it occurs, is exactly the failure of clause (ii) for every window-surviving blind control: the post-observation recourse mode.

Complete proof of the window-measurability no-go (main text, Section 7). The instance of Supplementary S3/A.3 has states x_1, x_2, x_3, x_4 with safe set $\{x_1, x_2, x_4\}$, transitions $x_1: a \rightarrow x_1, b \rightarrow x_4$; $x_2: a \rightarrow x_4, b \rightarrow x_2$; $x_4: a \rightarrow x_3, b \rightarrow x_3$; initial belief $B_0 = \{x_1, x_2\}$, blind at the first step and exactly informed after. The variant changes only x_4 's post-reveal transitions to $a \rightarrow x_4, b \rightarrow x_4$. All window sub-model data coincide: the same B_0 , the same first-step transitions of x_1 and x_2 , the same floor, the same schedule. Both certificates are silent in each: each blind action keeps both branches in the safe set through the window step, so the common-action and timing certificates have no witness. The exact recursion decides differently. In the instance, every post-reveal action from x_4 exits the safe set, so $x_4 \notin \text{RViab}(\mathcal{V})$, the one-step-backward pass misses B_0 at horizon two, and B_0 is nonviable; in the variant, x_4 is

maintained under both actions, $x_4 \in \text{RViab}(\mathcal{V})$, the action a lands x_1 and x_4 in the kernel, and B_0 is viable. Any window-measurable pair returns identical verdicts on the two instances, so no window-measurable pair is complete. The verification record confirms each step in exact arithmetic: certificate silence in both instances; nonviability of the instance at horizon two with failure exactly on clause (ii) of the decomposition (every window-surviving blind action lands a branch outside the kernel); and viability of the variant.

S3. Bounded constructions and scope remarks

Appendix A: Bounded Constructions and Scope Remarks

The bounded constructions below complement the obstruction theorems of Sections 3–4. They are stated in full, together with their scope remarks.

A.1 Example (coupling creates viability absent in a factor). Take the two-patch logistic system $\dot{S}_i = g_i(S_i) + \kappa(S_j - S_i) - h_i$, $i, j \in \{1, 2\}$, $i \neq j$, with $g_i(s) = s(1 - s)$, $\kappa = 0.2$, constraints $S_i \in [0, 1]$ and $h_i \geq h_{\min, i}$ (harvest floors), and admissible controls the constant harvest vectors $h = (h_1, h_2)$ with $h_i \geq 0$. Choose $(S_1^*, S_2^*) = (0.5, 0.8)$. Define harvest floors by the equilibrium equations: $h_{\min, 1} = g_1(0.5) + 0.2(0.8 - 0.5) = 0.31$; $h_{\min, 2} = g_2(0.8) + 0.2(0.5 - 0.8) = 0.10$. Patch 1 in isolation: $\max_s g_1(s) = 0.25 < h_{\min, 1} = 0.31$, so patch 1 is not viable in isolation. Yet the coupled system has the equilibrium $(0.5, 0.8)$ with $h = (0.31, 0.10)$ admissible and the stock constraints satisfied, so its kernel is nonempty. The example is the constructive counterpart of the emptiness results of Section 3: where the obstruction theorems certify that no policy works for reasons of information, patch coupling exhibits the opposite direction — a jointly viable operating point that no factor sustains alone.

A.2 Counterexample (emptiness despite factorwise viability at MSY). Take $\kappa > 0$, $C_1 \neq C_2$, $h_{\min, i} = r_i C_i / 4$ (MSY level), in the same coupled system with $\phi_i(S_i) := g_i(S_i) - h_{\min, i}$ and growth laws $g_i(S) = r_i S(1 - S/C_i)$, so that

$$\phi_i(S_i) = -\frac{r_i}{C_i} \left(S_i - \frac{C_i}{2} \right)^2 \leq 0$$

with equality only at $S_i = C_i/2$, and the constraint set is the box $[C_1/2, \bar{S}_1] \times [C_2/2, \bar{S}_2]$. Each isolated system has kernel $[C_i/2, \bar{S}_i]$ (from $S_i > C_i/2$ the stock declines toward $C_i/2$ asymptotically, never below). For the coupled system, define the Lyapunov function $W = S_1 + S_2$; then

$$\dot{W} = \phi_1(S_1) + \phi_2(S_2) \leq 0,$$

with equality exactly at $p^* = (C_1/2, C_2/2)$, where the flow is nevertheless nonzero: $(\dot{S}_1, \dot{S}_2)|_{p^*} = (\frac{\kappa}{2}(C_2 - C_1), \frac{\kappa}{2}(C_1 - C_2)) \neq 0$. Suppose a trajectory remained in the constraint box for all time. Then W is non-increasing and bounded below by $(C_1 + C_2)/2$, so $W(t) \downarrow W_\infty \geq (C_1 + C_2)/2$; the ω -limit set of the trajectory is nonempty, compact, and invariant, and $\dot{W} \equiv 0$ on it (otherwise W would keep decreasing along the set). Hence the ω -limit set is contained in $\{p^*\}$, so it equals $\{p^*\}$ and the trajectory converges to p^* . But convergence to p^* is impossible: along a trajectory with $x(t) \rightarrow p^*$ the velocity satisfies $\dot{x}(t) = f(x(t)) \rightarrow f(p^*) \neq 0$, so $x(t) - p^* \sim (t - t_0)f(p^*)$, contradicting convergence. Therefore no trajectory remains admissible for all time: the kernel is empty, and since the box is compact and the vector field smooth, every trajectory violates the constraint in finite time. Note the logical order: the emptiness conclusion here rests on the Lyapunov– ω -limit argument, *not* on the absence of an equilibrium — absence of equilibria alone does not imply an empty kernel (periodic and nonstationary viable trajectories are possible in general), and the two are kept separate. The counterexample is specific to the MSY parameter choice; for $H_{\min, i} < r_i C_i / 4$, equilibria may exist. Together with A.1 it delimits what coupling can and cannot repair: factorwise nonviability is repairable (A.1), but factorwise viability at an incompatible operating point is not (A.2).

A.3 Example (insufficient post-observation recourse). A three-state system shows the mode of Section 6.5(ii) concretely. Take $\mathcal{V} = \{x_1, x_2, x_4\}$ with $x_3 \notin \mathcal{V}$ a violation, actions $A = \{a, b\}$, and no disturbance. Transitions: x_1 maps to x_1 under a and to x_4 under b ; x_2 maps to x_4 under a and to x_2 under b ; x_4 maps to x_3 under both actions, so x_4 lies in $\mathcal{V}$ but outside the kernel. The observation merges x_1 and x_2 at the first step and is exact thereafter. The root belief $B_0 = \{x_1, x_2\}$ is one-step viable: action a keeps both states in $\mathcal{V}$ (reaching x_1 and x_4), as does b (reaching x_4 and x_2), so the common-action certificate is silent and no exit occurs in the first step. After the first observation, however, the trajectory is at x_4 for one of the two possible initial states (under a it is x_2 , under b it is x_1), and x_4 has no safe action, so that branch exits at the second step regardless of what the policy learns. Hence $B_0 \notin \mathcal{W}_2$ although $B_0 \in \mathcal{W}_1$: the failure is a post-observation recourse failure, witnessed only by the recursion (Theorem 1) — the exit certificate is silent at the root because both x_1 and x_2 are individually viable under full information, and the timing certificate is silent because the observation arrives without delay. The main text supplies the continuous-time counterpart (Section 3.7 there): an oracle-relaxed recourse obstruction whose three-branch braking instance fires exactly in this regime — blind-window safety with insufficient post-observation braking authority — with exact delay threshold $\tau = 7/50$.

S4. Additional figures

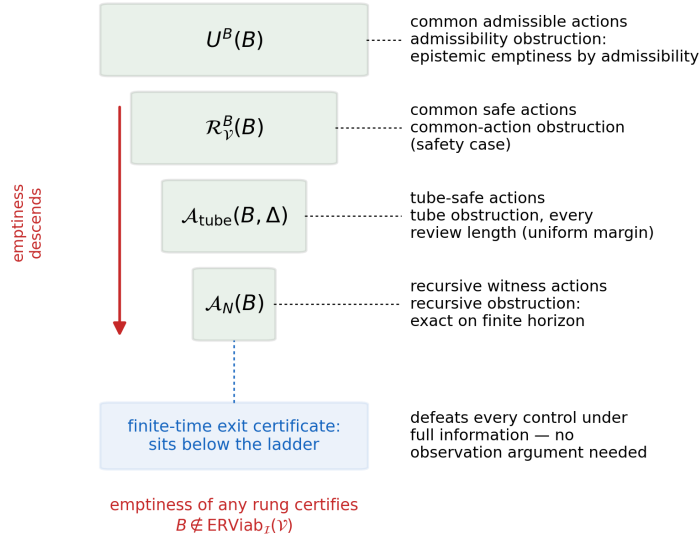


Figure 2: The obstruction ladder. The common response sets form nested necessary conditions for viability, $U^B(B) \supseteq \mathcal{R}_V^B(B) \supseteq \mathcal{A}_{\text{tube}}(B, \Delta) \supseteq \mathcal{A}_N(B)$, and emptiness descends (Proposition 2); the finite-time exit certificate sits below the ladder, defeating every control under full information (Theorem 4).

Auxiliary definitions and remarks (for reference)

Definition 1 (robust epistemic kernel). $B_0 \in \text{ERViab}_T(\mathcal{V})$ if there exists a nonanticipative observation-based policy π such that, for every compatible initial state $x_0 \in B_0$ and every admissible disturbance realization $d(\cdot)$, the trajectory of $\dot{x} = f(x, u(t), d(t))$ generated from x_0 by π 's actions under $d(\cdot)$ — the record being the one that $(\pi, d(\cdot))$ generates — remains in $\mathcal{V}$ for all time. (This is the standard exists-strategy-for-all-realizations form. The quantifier order is essential: the record is generated by the realization, so conditioning the admissible realizations

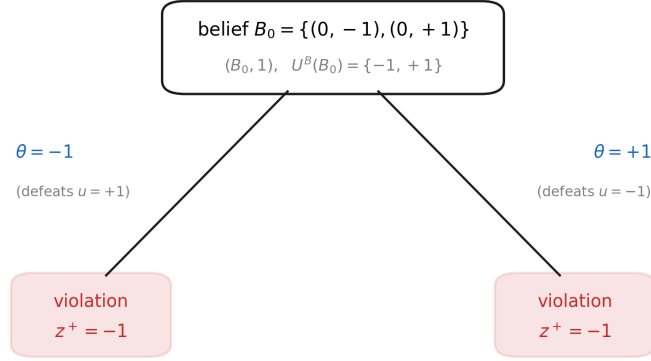


Figure 3: The one-step obstruction tree of Example 1, read as a finite system: for each admissible action the adversary has a compatible branch terminating in violation, so $B_0 \notin \mathcal{W}_1$ (Theorem 1).

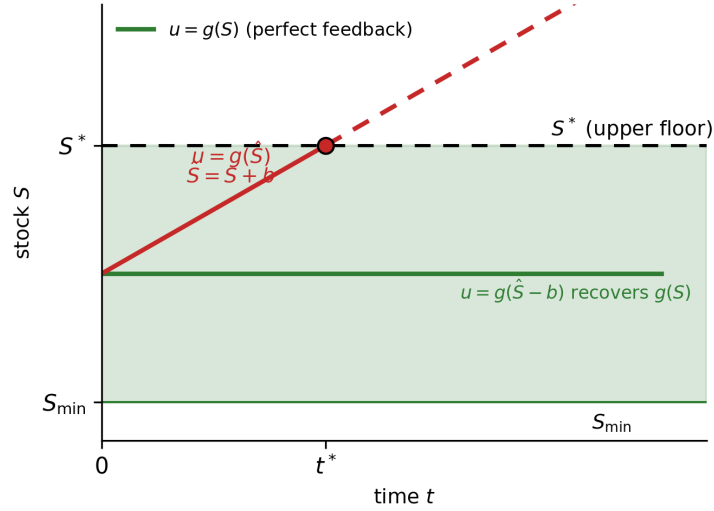


Figure 4: The certainty-equivalence trap. Under the perfect-information law $u = g(S)$ the stock is stationary inside $\mathcal{V}$. Applying the same law to the biased observation $u = g(\hat{S})$ with $\hat{S} = S + b$ gives $\dot{S} = g(S + b) - g(S) > 0$, so S exits above S^* in finite time (Remark 3); inverting the bias restores $g(S)$.

on the record would be circular.) Admissible initial beliefs are the compact information sets $B_0 \subseteq X$ generated by the observation structure — the observation fibres $O^{-1}(y)$, $y \in O(X)$, in the constructions of Sections 3.2–3.3 and 3.6. The existential (non-robust) counterpart $\text{EViab}_{\mathcal{I}}(\mathcal{V})$ is the non-robust contrast class.

Proposition 6 (selector principle). If $B \in \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ and π is a policy witnessing viability on B , then at every review time the action $a = \pi(B)$ selected by π is a response that is (i) *admissible*, $a \in U^B(B)$; (ii) *safe*, keeping every compatible trajectory in $\mathcal{V}$ until the next observation — $\text{Reach}_{[0, \Delta]}(B, a) \subseteq \mathcal{V}$ at review length Δ ; and (iii) *recursively viable*, in that every post-observation belief reached at the next review is again in $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$ under the continuation of π . A belief is therefore viable only if each of the response correspondences of this section — $U^B(B)$, $\mathcal{R}_{\mathcal{V}}^B(B)$, $\mathcal{A}_{\text{tube}}(B, \Delta)$, and the recursive predecessor of Section 3.1 — is nonempty along every reachable branch; hence a certificate that any one of them is empty at a reachable belief certifies nonviability. In finite systems the converse is exact (Theorem 1); in continuous time the converse is the backward-recursion completion noted in Section 6.5.

Remark 1 (comparison-function form). Hypothesis (H5.1) uses a constant drift margin ε . The certificate extends to state-dependent decline: if there is a continuous $\alpha : (0, a] \rightarrow (0, \infty)$ with

$$\sup_{u \in U(x)} \inf_{d \in D(x)} D^+ q(x; f(x, u, d)) \leq -\alpha(q(x)) \quad \forall x \in \mathcal{S}_a,$$

then the exit-time bound sharpens to

$$t_{\text{exit}} \leq \int_0^{q(x_0)} \frac{ds}{\alpha(s)},$$

whenever the integral is finite. The proof replaces the linear comparison $q(x(t)) \leq q(x_0) - \varepsilon t$ with the solution r of $\dot{r} = -\alpha(r)$, $r(0) = q(x_0)$, by the standard comparison argument. If $\alpha(0) = 0$ and the integral diverges, the conclusion weakens to asymptotic approach rather than finite-time exit; the constant-margin case $\alpha \equiv \varepsilon$ is the sharp finite-time instance.

Example 1 (hidden-mode conflict). Let an unobserved parameter satisfy $\theta \in \{-1, +1\}$, with $\dot{z} = \theta u$, $u \in \{-1, +1\}$, and constraint $z \geq 0$. At $z = 0$: if $\theta = +1$, only $u = +1$ is safe; if $\theta = -1$, only $u = -1$ is safe. Both compatible states are individually robustly viable under full information (each admits its safe control), but the information set $B = \{(0, +1), (0, -1)\}$ satisfies $\mathcal{R}_{\mathcal{V}}^B(B) = \{+1\} \cap \{-1\} = \emptyset$, so $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$. This is a purely informational failure: no stochasticity and no estimation-quality argument is involved. The example is the two-action skeleton of every “the stock is either recovering or collapsing, and the safe policy differs between the two” situation.

Remark 2 (threshold form of the timing obstruction). The delayed-information obstruction has a sharp threshold. Let $\sigma^*(B_0)$ be the guaranteed blind-window survival time: the largest duration such that some pre-observation policy keeps every observation-equivalent branch in $\mathcal{V}$ until then,

$$\sigma^*(B_0) = \sup_{u(\cdot)} \inf_{x_0 \in B_0, d(\cdot)} \tau(x_0, u, d),$$

with the infimum over branches that remain observation-equivalent on $[0, T_{\text{obs}})$ and τ the first violation time under the convention of Theorem 3. Then

$$\sigma^*(B_0) < T_{\text{obs}} \implies B_0 \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V}),$$

because before T_{obs} a policy cannot condition its action on which branch realizes. Condition (3) certifies this threshold: under (H3.1)–(H3.2) every branch exits by time $\inf_{x \in B_0} q(x)/\varepsilon$, hence $\sigma^*(B_0) \leq \inf_{x \in B_0} q(x)/\varepsilon$, and (3) states that this upper bound lies below T_{obs} ; the threshold itself may be strictly smaller, in which case nonviability holds even when (3) fails. The threshold is the blind-window response correspondence of the timing layer: its emptiness is the timing certificate, the delayed-information analogue of the tube and common-action correspondences.

Definition 2 (exact certifier). For an admissible domain $Z \subseteq X$ and a safe set $K \subseteq Z$, an **exact certifier** based on the observation map $O : Z \rightarrow Y$ is a function $C : Y \rightarrow \{0, 1\}$ such that

$$C(O(z)) = 1 \iff z \in K \quad \text{for every } z \in Z.$$

Remark 3 (the certainty-equivalence trap). Consider $\dot{S} = u - g(S)$ with $u \in [0, \bar{u}]$, g strictly increasing on $[S_{\min}, S^* + b]$, $g(0) = 0$, and $\mathcal{V} = [S_{\min}, S^*]$. Assume the range condition $g(S + b) \leq \bar{u}$ for every $S \in [S_{\min}, S^*]$, so that the controller below is admissible. Under perfect information the zeroing feedback $u(t) = g(S(t))$ gives $\dot{S} = 0$, so every $S_0 \in \mathcal{V}$ is viable and $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}}) = \mathcal{V} \neq \emptyset$. Now take the injective, biased observation $\hat{S} = S + b$ with $b > 0$, and the canonical certainty-equivalence controller: the perfect-information zeroing law applied directly to the uncorrected observation, $u = g(\hat{S}) = g(S + b)$. Then

$$\dot{S} = g(S + b) - g(S) > 0 \quad \forall S,$$

and since g is strictly increasing on the compact interval $[S_{\min}, S^* + b]$, $\dot{S}$ is bounded below by a positive constant there; hence S strictly increases and exits above S^* in finite time from every $S_0 \in \mathcal{V}$: the kernel of this single controller is empty. The bias-corrected controller $u = g(\hat{S} - b) = g(S)$ uses the observation map's structure, restores $\dot{S} = 0$, and keeps the kernel nonempty.

The mechanism is a *failure of one policy to use the observation map's structure*, not a loss of information: for a known invertible observation the output-feedback class is as powerful as the state-feedback class. The phenomenon is the viability-theoretic analogue of Witsenhausen's counterexample in stochastic control (Witsenhausen, 1968): the information pattern — not the dynamics and not the constraint — defeats the uncorrected controller. The remark is policy-specific; it does not assert that the entire certainty-equivalence class empties the kernel.

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